短时傅里叶变换 STFT

$f \in L^2(\mathbb{R})$ 时频型变换：F 有 τ, ω 两个参数

FT 则要求 $f \in L^1(\mathbb{R})$ 更严格

窗函数 $g(t)$： Rectangle. Gaussian. Hamming

窗越宽，频域分辨率越高；窗越窄，时域分辨率越高

基 $\{g_{\tau,\omega}\}$ $\quad g_{\tau,\omega}(t) = g(t-\tau)e^{j\omega t}$

STFT $\quad F(\tau,\omega) = Sf = \int f(t) g^*_{\tau,\omega}(t)\,dt = \int f(t) g^*(t-\tau) e^{-j\omega t}\,dt$

逆变换 $\quad f(t) = \frac{1}{2\pi}\iint F(\tau,\omega) g_{\tau,\omega}(t)\,d\tau\,d\omega$

[例] $f(t) = e^{j\omega_0 t} \qquad F(\omega) = \delta(\omega-\omega_0)$ 频移性质

$F(\tau,\omega) = \int f(t) g(t-\tau) e^{-j\omega t}\,dt = \delta(\omega-\omega_0) e^{j\omega(t-\tau)}$ 时移性质

Parseval 等式

$\int |f(t)|^2 dt = \frac{1}{2\pi}\iint |F(\tau,\omega)|^2 d\tau\,d\omega$ $\qquad$ FT $\int |f(t)|^2 dt = \frac{1}{2\pi}\int |F(\omega)|^2 d\omega$

Pf: $\int |f(t)|^2 dt = \int f(t)[\mathcal{F}^{-1}F(\omega)]^* dt = \int f(t)\int F^*(\omega) e^{j\omega t}\,d\omega\,dt$ 交换顺序

$= \int F^*(\omega)\underbrace{\int f(t) e^{j\omega t}\,dt}_{=\frac{1}{2\pi}F(\omega)\ ?}\,d\omega = \int F^*(\omega)\cdot\frac{1}{2\pi}F(\omega)\,d\omega$

什么样的 $F(\tau,\omega) \in L^2(\mathbb{R}^2)$ 可当作变换

核 $K(\tau,\omega,\tau_0,\omega_0) = \langle g_{\tau,\omega}, g_{\tau_0,\omega_0}\rangle$

$F(\tau_0,\omega_0) = \frac{1}{2\pi}\iint F(\tau,\omega) K(\tau,\omega,\tau_0,\omega_0)\,d\tau\,d\omega$

[例] Gabor 小波 $\psi(t) = \underbrace{\frac{1}{(\sigma^2\pi)^{1/4}} e^{-\frac{t^2}{2\sigma^2}}}_{g(t)\ 高斯} e^{j\omega_0 t}$

$\hat{g}(\omega) = (4\pi\sigma^2)^{1/4} e^{-\frac{1}{2}\sigma^2\omega^2}$

$\psi(t) = \frac{1}{\sqrt[4]{\pi}} e^{-\frac{1}{2}t^2} e^{j\omega_0 t}$

$\|\psi(t)\|_2^2 = \frac{1}{\sqrt{\pi}}\int |e^{-\frac{1}{2}t^2} e^{j\omega_0 t}|^2 dt$

$= \frac{1}{\sqrt{\pi}}\int |e^{-\frac{1}{2}t^2}|^2 \cdot \cancel{|e^{j\omega_0 t}|^2}\,dt = \frac{1}{\sqrt{\pi}}\int e^{-t^2} dt = 1$

$\psi_{\tau,f}(t) = \frac{1}{\sqrt[4]{\pi}} e^{-\frac{1}{2}(t-\tau)^2} e^{j2\pi f t}$

离散

基 $g_{\tau,m}[n] = g[n-\tau]e^{j2\pi mn/N}$

DSTFT $F[\tau,m] = Sf = \sum_{n=0}^{N-1} f[n]g^*[n-\tau]e^{-j2\pi mn/N}$

Inverse $f[n] = \frac{1}{N}\sum_{\tau=0}^{N-1}\sum_{m=0}^{N-1} F[\tau,m]g[n-\tau]e^{j2\pi mn/N}$

Parseval $\sum_{n=0}^{N-1}|f[n]|^2 = \frac{1}{N}\sum_{\tau=0}^{N-1}\sum_{m=0}^{N-1}|F[\tau,m]|^2$

小波变换 Wavelet

母小波 $\psi(t)$ 长度必需有限

要求：$E\psi(t)=0$，且 $C_\psi = \int \frac{1}{\omega}|\hat{\psi}(\omega)|^2 d\omega < +\infty$

基 $\{\psi_{\tau,s}\}$ $\psi_{\tau,s}(t) = \frac{1}{\sqrt{s}}\psi(\frac{t-\tau}{s})$ s 为放缩倍率

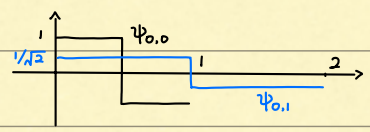
小波变换 $F(\tau,\omega) = Wf = \int f(t)\psi^*_{\tau,s}(t)dt = \int f(t)\frac{1}{s^2}\psi^*(\frac{t-\tau}{s})dt$

Inverse $f(t) = \frac{1}{C_\psi}\int_0^\infty\int_{-\infty}^\infty F(\tau,\omega)\psi_{\tau,\omega}(t)d\tau\frac{1}{s^2}ds$

Parseval 等式 $\int|f(t)|^2dt = \frac{1}{C_\psi}\int_0^\infty\int_{-\infty}^\infty|F(\tau,\omega)|^2d\tau\frac{1}{s^2}ds$

[例] Haar 母小波 $\psi(t)$

$\psi_{\tau,s}(t) = (\frac{1}{\sqrt{2}})^s\psi(\frac{t-\tau}{2^s})$



离散 Harr 变换

[例] $x = [8\ 4\ 1\ 3]$

求均值 $[\underbrace{8\ 4}_{6}\ \underbrace{1\ 3}_{2}]$

求差值 $[\underbrace{8\ 4}_{2}\ \underbrace{1\ 3}_{-1}]$

求均值 $[\underbrace{6\ 2}_{4}]$

求差值 $[\underbrace{6\ 2}_{2}]$

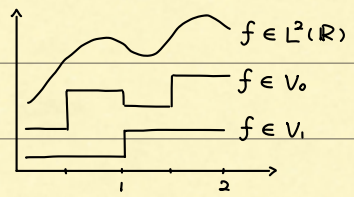
$\hat{x} = [4\ 2\ 2\ -1]$

Harr 矩阵 $\hat{x} = Wx$ $W = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 2 & -2 & 0 & 0 \\ 0 & 0 & 2 & -2 \end{pmatrix}$

2D-Harr 小波

超分辨分析

V_{s_0}：最细格点为 $s_0/2$ 的函数空间



$$\underbrace{V_\infty}_{=\{f(t)=0\}} \subset \cdots \subset V_1 \subset V_0 \subset \cdots \subset \underbrace{V_{-\infty}}_{=L^2(\mathbb{R})}$$

W_{s_0}：正交基 $\{\psi_{\tau,s_0}(t),\ \tau \in \mathbb{Z}\}$ 函数空间

$$W_{s_0} = V_{s_0}^{\perp}, \quad V_{s_0-1} = V_{s_0} \oplus W_{s_0}$$

故 $\{\psi_{\tau,s}\}$ 为 $L^2(\mathbb{R})$ 的正交基 $\quad L^2(\mathbb{R}) = \bigoplus_{s=-\infty}^{\infty} W_{s_0}$